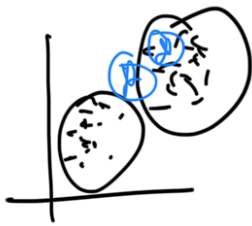


COL 774 / COL 734
Nov 4 (Evening) 2015



$$K=2 \quad \{x^i\}_{i=1}^m$$

$\{c^i\}_{i=1}^m$:- cluster assignments

$\{u_k\}_{k=1}^K$:- cluster means

GMM: Gaussian Mixture Model

$$\{x^i, y^i\}_{i=1}^m$$

$$\theta = (\{u_k, \Sigma_k, \pi_k\})$$

Recall: GDA:

$y^i \sim \text{Multinomial}(\Phi)$

$x^i | y^i = k \sim N(\mu_k, \Sigma_k)$

GMM :-

$t \leftarrow 0$

$\theta(t) \leftarrow \text{init}()$

do

E-step

$$z^i(t+1) \leftarrow \underset{z}{\text{argmax}} [p(z^i | x^i; \theta(t))]$$

M-step

$$\theta(t+1) \leftarrow \underset{\theta}{\text{argmax}} [p(\prod_{i=1}^m (x^i, z^i(t+1); \theta))]$$

$t \leftarrow t+1$
(while (converged))

E-step :-

$$p(z^i | x^i; \theta) =$$

$$\frac{p(x^i | z^i; \theta) p(z^i; \theta)}{\sum_j p(x^i; \theta)}$$

+ constant

M-step :

$$\mu_k = \frac{\sum_{i=1}^m 1\{z^i = k\} x^i}{\sum_{i=1}^m 1\{z^i = k\}}$$

$$\pi_k \equiv p(z^i = k | x^i; \theta)$$

$$\mu_k =$$

$$\frac{\sum_{i=1}^m 1\{z^i = k\} x^i}{\sum_{i=1}^m 1\{z^i = k\}}$$

$$\hat{\Sigma}_k = \frac{\sum_{i=1}^m \{z^u_i = k\} (x^u_i - \mu_k)(x^u_i - \mu_k)^T}{\sum_{i=1}^m \{z^u_i = k\}}$$

GMM

```

t ← 0;
Θ(0) ← init();
do
    zu(t+1) ← p(zu | xu; Θ(t))
    Θ(t+1) ← argmaxΘ ∏i=1m p(xui, zui; Θ)
    t ← t+1;
} while ! converged

```

k-Means

```

t ← 0;
μ(0) ← init();
do
    zu(t+1) ← argminz J(z, μ(t))
    μ(t+1) ← argminμ J(zu(t+1), μ);
    t ← t+1;
} while ! converged

```

(Try the 2 examples done in
 using class pathological cases for k-Means)

⇒ EM:-

↳ Expectation Maximization :-

$$\{x^u\}_{i=1}^m$$

$$x^u_i \in \mathbb{R}^n$$

$$L(\theta) = \log \left[\prod_{i=1}^m p(x^u_i; \theta) \right]$$

$$= \log \prod_{i=1}^m \sum_{z^u} p(x^u_i, z^u; \theta)$$

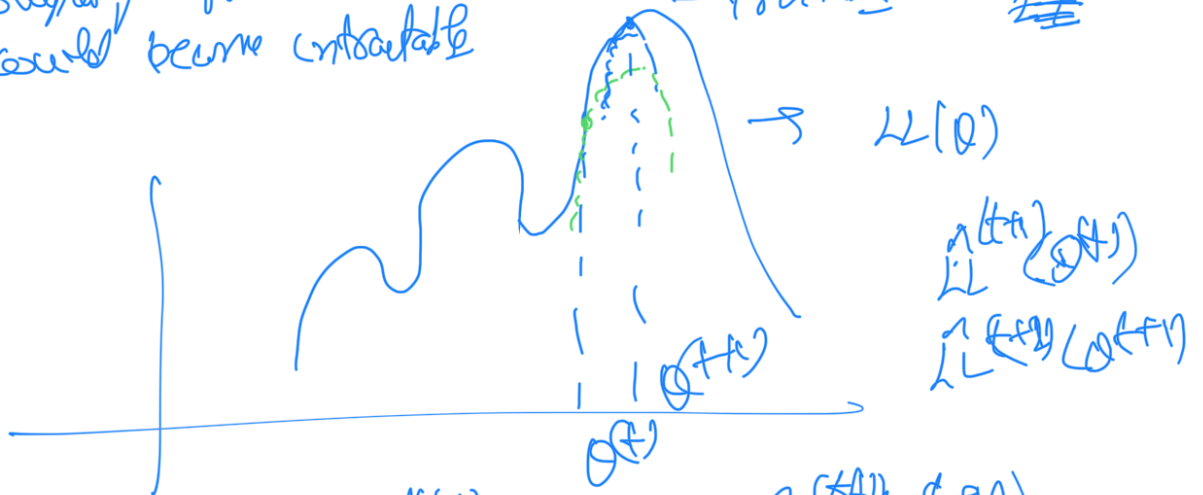
$$(x^u_i, z^u_i) \sim \text{Dist}(\theta)$$

$$p(x^u_i, z^u_i; \theta)$$

$$\argmax_{\theta} \log \prod_{i=1}^m p(x^u_i, z^u_i; \theta)$$

↳ computing gradient
could become unstable

↳ is tractable ~~is~~



$$f(\theta) \leq g(\theta)$$

then-
if $f(\theta^{t+1}) = g(\theta^{t+1})$

$$\Rightarrow \nabla_{\theta} f(\theta) |_{\theta^{t+1}} = \nabla_{\theta} g(\theta^{t+1})$$

$$\begin{aligned} \nabla_{\theta} L^{(t+1)}(\theta^{t+1}) &\leq \nabla_{\theta} L^{(t+1)}(\theta^{t+1}) \\ &\leq \nabla_{\theta} L(\theta^{t+1}) \\ &= \nabla_{\theta} L^{(t+1)}(\theta^{t+1}) \end{aligned}$$